



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VI Semester
End Semester Examination - April / May -2025
Low Power VLSI Design [ECE-4063]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) The chip size of a CPU is 10mm*10mm with a clock frequency of 350 Mhz operating at 5V. The length of the clock routing is estimated to be twice of the circumference of the chip. Assume that the clock signals routed on a metal layer with a width of 1.5 micro metre and parasitic capacitance of metal layer is 1Ff/ micro meter. Estimate the power dissipation of the clocked signal. (4)
 - A)
 - B) Calculate the switching frequency for a half adder circuit (Sum & Carry) assume the time period is 10ns for each bit input. (3)
 - C) Describe why low power consumption is essential in VLSI design (3)
- 2) Determine the static signal probability for 3 Bit Binary to Gray code converter using basic gates (4)
 - A)
 - B) Find the activity factor for the function $F=A \text{ XOR } B \text{ XOR } C$ in basic gates. Assume $P(A)=P(B)=P(C)=0.5$ (3)
 - C) Define gate level power analysis using transition density (3)
- 3) Given the switching function $F(A,B,C)=(A+BC+B'C)'$. Assume that all inputs toggle with a probability of 0.5. Calculate the activity factor for all intermediate outputs and outputs. Assume $\alpha=1$. (4)
 - A)
 - B) Consider the counter that counts in the following sequence such as 0,7,1,6,5 or 0,1,3,2,4. Determine which sequence has minimum power dissipation. (3)
 - C) Calculate signal entropy for the function $F=AB+C$ for all the input, output and intermediate nodes. (3)
- 4) Represent $F=AB+BC+CA$ in reduced ordered binary decision diagram. Using Thornton and Nair's top down approach algorithm, compute the output probability for the ROBDD. (4)
 - A)
 - B) Consider the state machine encoding for a two bit odd/ even counter. If control signal input is 0, it behaves as an odd counter else it behaves as even counter. Compute the number of transitions and calculate $E(M)$. (3)
 - C) Explain state machine coding and transition analysis coding. (3)

- 5) Consider 4:1 multiplexer using 2:1 multiplexer and 4:1 multiplexer as a single block. Use technology mapping, estimate the mapping cost and area for both cases.
- A)

Standard cell	Area (μm^2)	Output capacitance (nF)	Input capacitance (nF)
4:1 multiplexer	300	0.3	0.3
2:1 multiplexer	190	0.2	0.02

(4)

- B) Military tower is sending the following set of serial data to the base station. It has been observed that a lot of power dissipation is taking place. Calculate the reduced number of transitions, power dissipation using SILENT coding technique on the data sent.

Name	T	T+1	T+2	T+3	T+4
D4	1	1	1	1	1
D3	0	0	0	0	0
D2	0	0	1	0	1
D1	0	1	0	0	0
D0	1	0	1	1	1

(3)

- C)

(3)

Determine the efficiency of coding for the following table using hamming distance bus invert method.

0	0	0	0	0	0	0	0
0	0	0	0	0	1	0	1
0	0	1	1	0	1	0	1
1	1	0	1	1	1	0	1
0	0	1	0	1	1	0	1
1	1	0	1	0	0	0	1
0	0	0	0	1	1	1	1
0	1	1	0	0	0	1	0
0	1	0	0	1	1	0	1
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0

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